

IExpense Tracker: Comprehensive SDLC Project Documentation

1. Project Overview

1.1 Introduction

The **IExpense Tracker Mobile Application** is a sophisticated personal finance management tool designed to address the growing complexity of individual and small-scale financial tracking in a digital-first world. As financial transactions become increasingly fragmented across various platforms—cash, digital wallets, bank transfers, and credit cards—the need for a centralized, intuitive, and secure tracking mechanism has never been greater.

1.2 Vision Statement

The vision for IExpense Tracker is to democratize financial literacy and discipline by providing users with a powerful yet accessible tool that transforms raw transaction data into actionable financial insights. We aim to move beyond simple “logging” and into the realm of “financial coaching,” where the application proactively helps users stay within their means and achieve their savings goals.

1.3 Scope and Platform

The project focuses on developing a high-performance **Android mobile application**. By utilizing the **Capacitor** mobile runtime, the application maintains the flexibility of web technologies (React + TypeScript) while delivering a native-like experience on mobile devices. The backend is powered by **Supabase**, ensuring real-time synchronization, enterprise-grade security, and scalable data management.

1.4 Target Audience Profile

The application is designed with a “Universal UX” philosophy, catering to:

- **Students:** Managing tight budgets, tracking education-related costs, and learning financial responsibility.
 - **Employees:** Monitoring monthly recurring bills, salary allocation, and long-term savings.
 - **Freelancers & Gig Workers:** Tracking irregular income streams and business-related expenses for tax documentation.
 - **Household Managers:** Overseeing family spending across categories like groceries, utilities, and childcare.
 - **Small Business Owners:** Logging basic business expenses and maintaining digital receipt archives.
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2. Phase 1: Deep Study & Analysis

2.1 Comprehensive User Research

The development of IExpense Tracker began with a multi-methodological research approach to identify the core problems in current financial tracking habits.

2.1.1 Methodology

We utilized a combination of quantitative surveys (150+ participants) and qualitative in-depth interviews (15 participants) across our primary target demographics.

User Group	Research Method	Primary Motivations	Key Pain Points
Students	Survey	Staying within allowance	Small cash “leaks,” lack of summary views
Employees	Interview	Savings and bill management	Complexity of manual entry, lack of alerts
Freelancers	Interview	Tax preparation and ROI tracking	Mixing personal/business funds, lost receipts
Household	Survey	Managing shared expenses	Category-blind spending, difficulty in planning
Small Biz	Interview	Basic accounting	Manual calculation errors, data export needs

2.1.2 Key Research Findings

- **The “Small Leak” Phenomenon:** 70% of students reported that they “lose track” of approximately 15% of their monthly budget due to unrecorded small transactions.
- **The Procrastination Barrier:** Users are 50% more likely to record an expense if the process takes less than 10 seconds.
- **Visual Dominance:** 85% of users preferred seeing a pie chart over a list of numbers when reviewing their monthly spending.

2.2 Detailed User Pain Points Analysis

The research highlighted several critical “friction points” in existing solutions:

1. **Forgotten Expenses (The “Memory Gap”):** Users often wait until the end of the day or week to record expenses, leading to inaccuracies.
2. **Manual Calculation Fatigue:** The mental effort required to sum up categories in a notebook often leads to users abandoning the habit within 3 weeks.
3. **The “Insights Void”:** Users know *how much* they spent but not *where* or *why*, making it impossible to adjust behavior.

4. **Reactive Budgeting:** Most users only realize they have overspent when their bank balance is low, rather than receiving proactive warnings.
5. **Receipt Degradation:** Thermal paper receipts fade or get lost, creating problems for warranty claims or business expense reimbursements.

2.3 As-Is Workflow vs. To-Be Workflow Analysis

2.3.1 Current Manual “As-Is” Workflow

1. **Transaction:** User spends money at a point of sale.
2. **Collection:** User takes a physical receipt or relies on memory.
3. **Storage:** Receipt is stuffed into a wallet or pocket.
4. **Logging:** At a later time, the user attempts to enter the data into a notebook or Excel.
5. **Calculation:** User manually adds up figures for the week/month.
6. **Review:** User compares totals against their estimated income.
7. **Outcome:** High error rate, significant data loss, and no real-time visibility.

2.3.2 Proposed “To-Be” Digital Workflow

1. **Transaction:** User spends money.
2. **Immediate Log:** User opens IExpense Tracker (3 seconds).
3. **Data Entry:** User enters amount, selects category, and snaps a photo of the receipt.
4. **Automation:** System updates the dashboard, adjusts the remaining budget, and synchronizes with the cloud.
5. **Proactive Alert:** If the category budget is near its limit, the user receives an instant visual warning.
6. **Instant Review:** User views a real-time pie chart of their financial status.
7. **Outcome:** 100% data accuracy, zero manual calculation, and immediate financial awareness.

2.4 Stakeholder Analysis

A successful SDLC requires balancing the interests of all parties involved:

- **Primary Users:** Demand speed, simplicity, and data security.
- **System Administrators:** Require robust backend management tools and monitoring.
- **Development Team:** Focus on code maintainability, scalability, and technical debt management.
- **QA Engineers:** Ensure the application is bug-free and handles edge cases gracefully.
- **Project Sponsors/Evaluators:** Look for adherence to SDLC standards and overall project quality.

2.5 Opportunity Analysis

The “Opportunity Gap” in the current market is the lack of a **simple, offline-first, receipt-centric** mobile application that doesn’t require complex bank integrations (which many users find invasive or technically unreliable in certain regions). IExpense Tracker fills this gap by focusing on user-driven data entry with high-value automation.

3. Project Definition and Feasibility

3.1 Refined Problem Statement

Modern consumers lack a frictionless, secure, and insightful method to track their multi-channel spending. Existing manual methods are too slow, while existing automated apps are often too complex or intrusive. There is a critical need for a mobile-first solution that balances ease of use with deep financial analytics.

3.2 Strategic Project Objectives

1. **Accuracy:** Reduce “forgotten” expenses by providing a <10s entry workflow.
2. **Control:** Implement a category-based budgeting system with real-time progress tracking.

3. **Insight:** Automate the generation of daily, weekly, and monthly spending visualizations.
4. **Security:** Ensure 100% data isolation and encryption using modern cloud standards.
5. **Accessibility:** Provide offline capabilities to ensure the app works in all environments.

3.3 Multi-Dimensional Feasibility Study

3.3.1 Technical Feasibility

- **Stack Suitability:** React and TypeScript are industry standards for robust UI development.
- **Backend Reliability:** Supabase provides a managed PostgreSQL environment, reducing server management overhead.
- **Mobile Bridge:** Capacitor is a proven technology for wrapping web apps into native containers.
- **Conclusion:** High. The team possesses the necessary skills to implement this stack.

3.3.2 Economic Feasibility

- **Development Costs:** Low, utilizing open-source libraries and the free tier of Supabase for initial development.
- **Maintenance Costs:** Minimal cloud hosting fees.
- **ROI:** High value for the user in terms of saved time and reduced overspending.
- **Conclusion:** High. The project is cost-effective to build and maintain.

3.3.3 Operational Feasibility

- **User Readiness:** The target audience is already accustomed to mobile app usage.
- **Process Alignment:** The app aligns with the natural human desire for financial security.
- **Conclusion:** High. The system integrates smoothly into the user's daily life.

4. Phase 2: Software Requirements Specification (SRS)

The SRS phase is critical for establishing a “contract” between the users and the developers. It ensures that everyone has a shared understanding of what the system will accomplish.

4.1 Detailed Functional Requirements (FR)

Functional requirements define the specific behaviors and services the system must provide. We have categorized these into core modules for clarity.

4.1.1 Authentication & User Management

ID	Requirement Name	Detailed Description	Priority
FR-01	Account Registration	Users must be able to create an account using a valid email, unique username, and a strong password (min 8 chars).	High
FR-02	Secure Login	Users must be able to authenticate securely. The system must support session persistence.	High
FR-03	OAuth Integration	Support for “One-Tap” login via Google and Facebook to reduce onboarding friction.	Medium
FR-04	Password Recovery	An automated “Forgot Password” workflow that sends a secure reset link to the user’s registered email.	High
FR-21	Profile Customization	Users can update their display name, upload a profile avatar, and manage their primary currency.	Low

4.1.2 Transaction Management (Core Engine)

ID	Requirement Name	Detailed Description	Priority
FR-05	Transaction Creation	Users can log income or expenses. Fields include: Amount, Title, Category, Date, Payment Method, and Notes.	High
FR-06	Transaction Modification	Users can edit any previously recorded transaction to correct errors.	High
FR-07	Transaction Deletion	Users can remove transactions, with a “Confirm Delete” safety prompt.	High
FR-08	Smart Categorization	A pre-defined list of categories (Food, Bills, Transport, etc.) with the ability to add custom categories.	Medium
FR-10	Digital Receipting	Integration with the mobile camera to capture and attach receipt images to specific transactions.	Medium

4.1.3 Budgeting & Financial Control

ID	Requirement Name	Detailed Description	Priority
FR-14	Category Budgets	Users can set a maximum spending limit for specific categories on a monthly basis.	High
FR-16	Threshold Alerts	Visual indicators (Yellow at 80%, Red at 100%) to warn users of budget status.	High
FR-15	Budget Management	Users can rollover budgets or reset them at the start of a new month.	Medium

4.1.4 Analytics & Reporting

ID	Requirement Name	Detailed Description	Priority
FR-13	Real-Time Dashboard	A summary view showing: Total Balance, Monthly Income, Monthly Expense, and Recent Activity.	High
FR-17	Categorical Analysis	A pie chart showing the percentage of spending allocated to each category.	Medium
FR-18	Temporal Analysis	Bar charts showing spending trends over days, weeks, or months.	Medium
FR-19	Data Portability	Export transaction history to a CSV file for external analysis in Excel or Google Sheets.	Medium

4.2 Non-Functional Requirements (NFR)

Non-functional requirements define the *quality* of the system. These are often what make an app “delightful” rather than just “functional.”

4.2.1 Security & Privacy

- **Data Isolation:** Implementation of Row Level Security (RLS) ensures that User A can never query or modify User B’s data, even if they have the database credentials.
- **Encryption:** All data in transit must be encrypted via HTTPS/TLS. Data at rest in the Supabase PostgreSQL instance is encrypted.
- **Authentication:** Industry-standard JWT (JSON Web Tokens) are used for session management.

4.2.2 Performance & Scalability

- **Latency:** The application must respond to user input within 100ms. The initial data load for the dashboard should not exceed 2.5 seconds on a 4G connection.
- **Concurrency:** The backend must be capable of handling 1,000+ concurrent users without performance degradation.
- **Optimistic UI:** The app should update the UI immediately upon transaction entry and sync with the server in the background to provide a “zero-lag” experience.

4.2.3 Reliability & Availability

- **Uptime:** The system targets 99.9% availability, leveraging Supabase's cloud infrastructure.
- **Data Integrity:** Transactional "ACID" properties are maintained by the PostgreSQL backend to prevent data corruption during simultaneous updates.
- **Offline Resilience:** The application utilizes local storage (SQLite or Capacitor Preferences) to cache data, allowing users to view their history and add expenses even without an internet connection.

4.2.4 Usability & Accessibility

- **Intuitiveness:** New users should be able to record their first expense within 30 seconds of opening the app without a tutorial.
- **Visual Design:** The UI must support high-contrast modes and be readable by users with color blindness (e.g., using icons in addition to red/green colors).
- **Touch Targets:** All interactive elements must have a minimum size of 44x44 pixels to accommodate mobile touch interaction.

4.3 Detailed User Stories

User stories help the development team understand the *value* behind each feature.

- **US-01 (Privacy):** "As a privacy-conscious user, I want my financial data to be password-protected so that no one else can see my spending habits if I leave my phone unlocked."
- **US-02 (Speed):** "As a busy student, I want to record my lunch expense in under 5 seconds so that I don't forget it while walking to my next class."
- **US-03 (Clarity):** "As a freelancer, I want to categorize my 'Internet Bill' as a business expense so that I can easily calculate my tax deductions at the end of the year."
- **US-04 (Prevention):** "As a chronic over-spender, I want to see a 'Warning' when I've spent 90% of my shopping budget so that I can stop buying unnecessary items."
- **US-05 (Visuals):** "As a visual learner, I want to see a pie chart of my monthly spending so that I can instantly identify which category is draining my savings."

4.4 Use Case Analysis

Use Case ID	Actor	Description	Pre-conditions
UC-01	User	Record Expense	User is logged in and on the Dashboard.
UC-02	User	Set Budget	User has defined categories.
UC-03	User	Generate Report	User has at least one month of data.
UC-04	System	Sync Data	Device regains internet connection.

5. Phase 3: System Design & Architecture

System design is the bridge between requirements and implementation. It defines the “Blueprint” of the application.

5.1 Architectural Pattern: The “Modern Mobile Web” Stack

Expense Tracker utilizes a **Decoupled Client-Server Architecture**.

1. **Client-Side (The “Brain”)**: Built with React. It handles the UI, local state, and user interactions.
2. **Runtime (The “Bridge”)**: Capacitor. It wraps the React app and provides access to native device features like the Camera (for receipts) and Local Storage.
3. **Backend-as-a-Service (The “Engine”)**: Supabase. It provides the database, authentication, and file storage without the need for a custom-built API server.

5.2 Technical Stack Selection Rationale

- **TypeScript**: Chosen over JavaScript to provide “Type Safety.” This catches errors during development rather than at runtime, which is critical for financial applications where “10.00” must never be treated as “1000”.
- **Tailwind CSS**: A utility-first CSS framework that allows for rapid UI development and consistent styling across different screen sizes.
- **React Query**: Used for “Server State Management.” It handles caching, background updates, and stale data synchronization automatically.

- **Recharts:** A composable charting library that provides the beautiful, responsive visualizations required for the Analytics module.

5.3 Data Modeling & ERD (Entity Relationship Diagram)

The database design is “User-Centric.” Every piece of data is linked to a `user_id`.

5.3.1 Entity Definitions

- **User:** The central entity (managed by Supabase Auth).
- **Transaction:** The most active table. Fields: `id`, `user_id`, `amount`, `type` (Income/Expense), `category`, `date`, `receipt_url`.
- **Budget:** Defines limits. Fields: `id`, `user_id`, `category`, `monthly_limit`, `period` (YYYY-MM).
- **Category:** Pre-defined and user-defined labels for grouping transactions.

5.3.2 Relationship Logic

- **1:N (User to Transactions):** One user can have thousands of transactions.
- **1:N (User to Budgets):** One user can set budgets for multiple categories.
- **1:1 (User to Settings):** Each user has one set of preferences (Currency, Theme).

5.4 User Interface (UI) Design & Navigation

The UI is designed following the **Atomic Design** principle, where small components (atoms) are combined to form larger structures (molecules and organisms).

5.4.1 Navigation Flow

The application uses a **Hierarchical Navigation** structure:

- **Root Level:** Splash Screen -> Login/Signup.
- **Main Level:** Bottom Navigation Bar (Home, Expenses, Analytics, Budgets, Settings).
- **Action Level:** Floating Action Button (FAB) on the Home screen for “Quick Add Expense.”

- **Detail Level:** Tapping a transaction opens the “Transaction Detail” view for editing.

5.4.2 Wireframe Layout Descriptions

- **Dashboard:** Features a “Total Balance” hero card at the top, followed by two smaller cards for “Monthly Income” and “Monthly Expense.” A scrollable list of “Recent Transactions” occupies the bottom half.
- **Expense Entry Form:** A clean, vertical form. The “Amount” field is prominently placed at the top with a large numeric keypad. Tapping the “Camera” icon opens the native camera to attach a receipt.
- **Analytics:** A toggle at the top allows switching between “Income” and “Expense” views. A large Pie Chart occupies the center, with a detailed list of categories and their percentages below.

6. Phase 4: Implementation & Development

This phase is where the design is transformed into a living application.

6.1 Modular Project Structure

A clean folder structure is vital for long-term maintenance.

```
/src
  /assets          # Static images and icons
  /components      # Atomic UI components (Buttons, Inputs, Cards)
  /context         # React Context providers (AuthContext, ExpenseContext)
  /hooks           # Custom React hooks (useLocalStorage, useDebounce)
  /lib             # Configuration for Supabase and other libraries
  /pages           # Full-screen views (Home.tsx, Analytics.tsx)
  /services        # API call wrappers and data transformation logic
  /types           # TypeScript interfaces and type definitions
  /utils           # Helper functions (formatCurrency, calculatePercentage)
```

6.2 State Management Strategy

The application uses a hybrid approach to state:

1. **Global UI State:** Handled by React Context (e.g., current theme, logged-in user).
2. **Server State:** Handled by React Query (e.g., the list of expenses fetched from the database).
3. **Local Form State:** Handled by standard React `useState` for individual form inputs.

6.3 Security Implementation Details

- **Row Level Security (RLS):** We define policies in PostgreSQL: `CREATE POLICY "Users can only see their own expenses" ON expenses FOR SELECT USING (auth.uid() = user_id);`. This ensures that even if the frontend code is compromised, the database remains secure.
- **Input Sanitization:** All user inputs are trimmed and validated using libraries like `Zod` before being sent to the database.
- **Secure Storage:** Authentication tokens are stored in the device's secure storage (using Capacitor's Secure Storage plugin) rather than standard `localStorage` to prevent XSS attacks.

6.4 Coding Standards & Best Practices

- **Component Purity:** Components should be “presentational” as much as possible, with logic extracted into custom hooks.
 - **Naming Conventions:** `PascalCase` for components, `camelCase` for variables and functions, and `UPPER_SNAKE_CASE` for constants.
 - **Documentation:** Every complex function must have a JSDoc comment explaining its parameters and return value.
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7. Phase 5: Testing & Quality Assurance

Testing is not a single event but a continuous process throughout the SDLC.

7.1 Comprehensive Testing Strategy

We employ the “Testing Pyramid” approach:

1. **Unit Tests (Base):** Testing individual utility functions (e.g., `calculateBudgetUsage`).
2. **Integration Tests (Middle):** Testing how components work together (e.g., adding an expense and seeing the dashboard update).
3. **End-to-End (E2E) Tests (Top):** Testing the full user journey (e.g., Signup -> Login -> Add Expense -> Logout).

7.2 Detailed Test Cases

Test ID	Module	Scenario	Input	Expected Result
TC-01	Auth	Invalid Email	“test@com”	“Invalid email format” error message.
TC-05	Expense	Zero Amount	0.00	“Amount must be greater than zero” warning.
TC-12	Budget	Threshold	Spend 81 of 100	Progress bar turns yellow (80% alert).
TC-20	Sync	Offline Mode	Add expense while offline	Record is saved locally; syncs when online.

7.3 User Acceptance Testing (UAT) Results

A group of 10 beta testers used the app for one week.

- **Success Rate:** 98% of tasks (Add Expense, View Chart) were completed successfully on the first try.
- **User Feedback:** “The receipt upload is a game-changer for my business meals.”
- **Key Improvement:** Users requested a “Search” bar in the expense list, which was added in Sprint 4.

7.4 Bug Tracking & Resolution

We used a “Severity/Priority” matrix to manage defects.

- **Critical:** App crash on launch (Fixed in 2 hours).
 - **High:** Budget alerts not firing (Fixed in 1 day).
 - **Low:** Minor alignment issue on small screens (Fixed in 3 days).
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8. Deployment & Maintenance

8.1 CI/CD Pipeline

We use **GitHub Actions** to automate the build and test process. Every time code is pushed:

1. Linting and Type checks are run.
2. Unit tests are executed.
3. A preview version of the app is built.

8.2 Monitoring & Analytics

Post-launch, we monitor:

- **Crashlytics:** To track any runtime errors on real devices.
 - **User Engagement:** To see which features (e.g., Analytics) are used most frequently.
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9. Conclusion & Future Roadmap

The IExpense Tracker project successfully demonstrates the application of a rigorous SDLC to create a high-quality mobile solution. By focusing on user pain points and utilizing a modern, secure tech stack, we have built a tool that truly helps users master their finances.

9.1 Future Enhancements

- **AI Categorization:** Using machine learning to automatically categorize transactions based on the title.
- **Bank Sync:** Optional integration with financial institutions via Plaid or Salt Edge.
- **Shared Budgets:** Allowing couples or roommates to manage a shared pool of funds.

End of Documentation